Pokémon Battle Prediction

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***Abstract*—We predict whether Player 1 wins a Gen 2 OU battle from Player 1's team, Player 2's lead Pokémon and the first 30 turns (10,000 training and 5,000 test battles). Our main pipeline reaches 0.690 stratified 5-fold cross-validation (CV) accuracy and 0.69666 public leaderboard accuracy; the team's best public accuracy is 0.69733.**

***Index Terms*—Pokémon battles, binary classification, feature engineering, ensemble learning, CatBoost, LightGBM, XGBoost.**

I. Introduction

The task is to predict the winner of a Gen 2 OU battle from partial battle information. Two properties guide the design: 69.5% of test Player 1 teams also occur in training, making random folds representative of the test distribution; and by turn 30, Player 2 has revealed about 5.4 of its 6 Pokémon.

II. Feature Engineering

Feature groups were added incrementally and evaluated with CV. Battle state—remaining HP per side (unseen Pokémon treated as full HP) and fainted counts—was strongest: logistic regression on these four features reached 0.654. Status-turn counts raised this to 0.658. Move usage, effects (confusion, Substitute, Leech Seed), HP lost/healed, switches, last-turn state and type-chart damage estimates (base power, STAB, effectiveness, stats, boosts, burn; best damage, turns to KO and speed order) brought CV to 0.686.

The final system combines two label-free feature sets. Set A ($\sim$1,670 numeric and 18 categorical columns) includes team base-stat aggregates, type matchups against the opposing lead, recency-weighted HP trajectories, boost/status statistics and a TF-IDF/SVD battle text view. Set B (333 columns, 8 categorical) is a compact set of engineered battle-state features. Label-based features are smoothed win rates (target encoding [3], smoothing 12–25) for 13 token groups (species, leads, matchups, moves), plus logistic and ridge probes on the TF-IDF matrix. They are recomputed with an inner 5-fold split within every training fold, so no validation label is used. Target encodings fitted without nesting inflated CV to about 0.70, while correctly nested encodings added only 0.1–0.2 points. LightGBM [2] ranks Set A columns within each training fold; the top 450 plus all aggregated state features are kept. Tested as separate groups, static team statistics, trend and move-habit features, and exact-team win rate (54% alone) did not improve CV beyond noise ($\approx$0.5 points).

III. Model Development

We began with regularized logistic regression and added CatBoost [4] and XGBoost [1]. On the same features, gradient-boosted trees performed on par with logistic regression (within about 0.3 points); gains came mainly from richer features and from ensembling models trained on different feature sets. In the main 5-fold pipeline, all settings were fixed in advance, without validation-fold tuning or early stopping. A larger CatBoost (1,500 iterations, depth 6), multi-seed averaging and a state-value model trained on 300,000 game states did not improve CV beyond noise.

Murad's standalone 10-fold multimodal ensemble combines CatBoost (depths 5–7), LightGBM and XGBoost with an internal training-set Pokédex, 18 tactical board-state categoricals, out-of-fold target encodings and TF-IDF/SVD linear probes, blended with simplex log-odds weights. The learned weights concentrated almost entirely on the CatBoost models (0.63, 0.33, 0.04). Unlike the main pipeline, it uses early stopping on each validation fold, optimized blend weights and a tuned decision threshold (0.489); it runs in about 115 minutes on Kaggle. It achieved the team's highest public score, 0.69733.

IV. Validation and Results

Primary choices used stratified 5-fold CV (seed 42). The public leaderboard covers only $\sim$1,500 test battles, so its estimated noise is $\pm$1.2 points; CV was therefore treated as the more reliable selection criterion. CV and public scores moved in the same direction, with public scores 0.5–1.3 points higher. A 5-fold StratifiedGroupKFold grouped by Player 1's six-Pokémon roster, combined with stripped mechanical features, achieved 0.6834 Group-OOF and 0.69333 public accuracy when blended.

Table 1. Stratified 5-fold CV and public leaderboard accuracy

Configuration (#features)	Model	CV	Public
Baseline (4)	LR	0.654	—
+ status (10)	LR	0.658	0.671
Core (112)	LR + CB + XGB	0.679	0.685
Core + matchup (169)	LR + 2×CB	0.680	0.682
Extended (276)	LR + CB	0.680	0.691
Extended + matchup (333)	LR + 2×CB	0.686	0.691
Final (Sets A+B)	LR + 3×CB	0.690	0.69666
Multimodal (473 + 18 cat.)	3×CB + LGBM + XGB	—*	0.69733

*The multimodal 10-fold OOF accuracy (0.724) used early stopping on validation folds, target encodings and feature pruning fitted on the full training set, and tuned weights and threshold; it is therefore optimistic and not comparable with the nested estimates above. LR: logistic regression; CB: CatBoost; XGB: XGBoost; LGBM: LightGBM.

The main pipeline averages four equally weighted models: three CatBoost models (800 iterations, learning rate 0.04, depth 5, L2 6) trained on Set A with label-based features, on pruned Set A with label-based features, and on Set B; plus a standardized logistic regression on Set B ($C = 0.03$). Feature extraction, 5-fold CV and final fitting took 755 s on Kaggle (CPU). Battle-state features provide most of the signal; all other groups add three to four points. Performance plateaus near 0.69 across very different feature sets and models, likely due to the 30-turn cutoff and battle randomness. Equal weighting reduces variance without extra tunable parameters, yielding 0.690 CV and 0.69666 public accuracy.

References

[1] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. KDD*, 2016.

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[4] L. Prokhorenkova *et al.*, "CatBoost: Unbiased boosting with categorical features," in *Proc. NeurIPS*, 2018.